Started on	Wednesday, 16 October 2024, 9:52 PM
State	Finished
Completed on	Sunday, 27 October 2024, 4:48 PM
Time taken	10 days 18 hours
Marks	1.00/1.00
Grade	**100.00** out of 100.00
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Question 1

Correct

Mark 1.00 out of 1.00

Write a Java Program to read n numbers and print the difference of a given number and 7.
Read position (another integer) and print the number in the given position - index starts from 0 for the given position.
Print "Invalid Position" when the position is not a valid position.
Print "Difference is Zero - Cannot Divide" when the difference is zero.

Handle appropriate exceptions.

Note: You can expect the difference to be sometimes zero.

For example:

Input	Result
5 13 8 9 10 11 3	6 1 2 3 4 3
5 9 8 7 10 11 2	2 1 0 3 4 Difference is Zero - Cannot Divide
3 7 7 8 7	0 0 1 Invalid Position

Answer: (penalty regime: 0 %)

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        try {
            int n = scanner.nextInt();
            int[] numbers = new int[n];
            for (int i = 0; i < n; i++) {
                numbers[i] = scanner.nextInt();
            }

            int[] differences = new int[n];
            for (int i = 0; i < n; i++) {
                differences[i] = Math.abs(numbers[i] - 7);
            }
            for (int diff : differences) {
                System.out.print(diff + " ");
            }
            System.out.println();
            int position = scanner.nextInt();

            if (position < 0 || position >= n) {
                System.out.println("Invalid Position");
            } else {
                int difference = numbers[position] - 7;
                if (difference == 0) {
                    System.out.println("Difference is Zero - Cannot Divide")
                } else {
                    System.out.println( difference);
                }
            }
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Invalid Position");
        } catch (Exception e) {
            System.out.println("An error occurred: " + e.getMessage());
        } finally {
```

```
            scanner.close();
        }
    }
}
```

	Input	Expected	Got	
✔	5 13 8 9 10 11 3	6 1 2 3 4 3	6 1 2 3 4 3	✔
✔	5 9 8 7 10 11 2	2 1 0 3 4 Difference is Zero - Cannot Divide	2 1 0 3 4 Difference is Zero - Cannot Divide	✔
✔	3 7 7 8 7	0 0 1 Invalid Position	0 0 1 Invalid Position	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.